



Dore ecology – EA data

A report prepared on behalf of Wye Salmon Association (WSA)

This report reviews the available EA ecological data relating to the Dore sub catchment. We review data for fish, invertebrates and macrophytes.

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Introduction

The motivation for this work is twofold. Firstly, to understand the data behind recent Water Framework Directive (WFD) classifications in the Dore sub catchment, and secondly to inform planning for potential WSA citizen science monitoring of invertebrates and macrophytes.

The Dore sub catchment is split into three waterbodies for the purposes of the WFD: the River Dore, the Dulas Brook and the Worm Brook. Recent classifications for these waterbodies are as follow:

Waterbody	2019			2022		
	Fish	Invert.	Macro.	Fish	Invert.	Macro.
River Dore	High	High	Moderate	Moderate	High	Moderate
Dulas Brook	High	High	Good	Moderate	High	Good
Worm Brook	Moderate	Good	Moderate	Moderate	Good	Moderate

Table 1: WFD ecological classifications for Dore sub catchment waterbodies

Fish numbers appear to have fallen precipitously on both the Dore and the Dulas. Meanwhile, invertebrates apparently remain at near-natural levels on both of these rivers and even achieve good status on the Worm. Only the Dulas gets a good for macrophytes, however. All of this begs the question: what are the numbers behind this? And are there any obvious long-term trends?

I have downloaded corresponding ecological data from the [EA Ecology and Fish Data Explorer](#), covering the period 1993 to 2023 for the Monnow operational catchment which incorporates the above waterbodies. The following sections separately examine the three ecological categories.

Fish

Fish numbers are monitored by electrofishing a section of river. This is generally done multiple times, until no further fish float to the surface (called a catch depletion sample). However, some surveys use only a single pass. Each species is counted and sometimes measured and weighed. The survey area depends on the width of the river and varies from year to year but is generally substantial; typical survey areas on the Dore, for example, are of the order of 500m². All surveys are performed in the summer months – typically July-September.

Over the period 1993-2023, surveys have been carried out at 12 different locations on the Dore, two on the Worm, and one on the Dulas. These are shown in figure 1 overleaf. However, examination of the timing (table 2 overleaf) shows a confusing picture with little overall structure or obvious logic. Two clusters on the Dore, in 1998 and 2017 appear to have been one-off spatial studies and included seven sites which have never been revisited.

Between 2003-19, the Dore at site 10542 (Poston Court Farm below Peterchurch) seems to have settled into a 5-year cycle. Indeed, this site is the only one in the sub catchment with a reasonable continuity of data. The follow-ups in 2021 & 22 are curious. Between 2013-19, site 50463 (Abbey Dore) was samples in alternate years, but this has not continued.

Dulas Brook has been surveyed only twice, at Ewyas Harold, and not since 2019. Which begs the question: what was the 2022 WFD classification for this waterbody based on?

Likewise, the Worm Brook has not been surveyed since 2013. So, on what basis was it classified in 2022, or 2019, or 2016? The answer would seem to be ‘old data’.

[illegible]

Table 2: fish surveys within the Dore sub catchment by site and year



Figure 1: map of the Dore sub catchment showing fish survey sites (orange circles).

Dore trends

In looking for long-term trends on the Dore, we really only have site 10542, Poston Court Farm, with any continuity. Between 1993 and the present, relatively few species have been recorded at this location: trout (*Salmo trutta*), and bullheads (*Cottus gobio*). Notable by their absence are eels, grayling, chub, minnows, lampreys & stone loach. Figure 2 below shows a normalised¹ trend chart for trout numbers and figure 3 shows the bullhead trend.

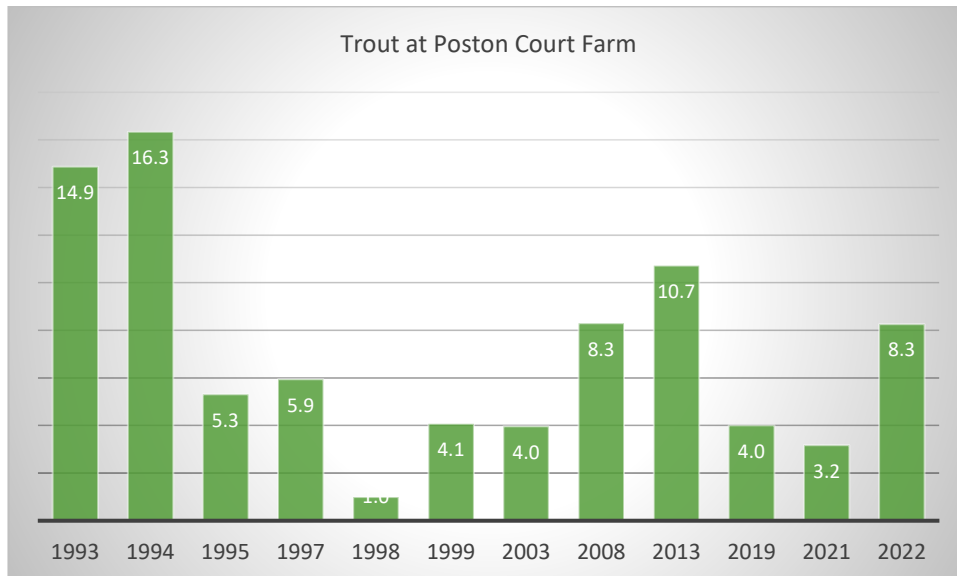


Figure 2: trout population (#/100m²) on the Dore at site 10542, Poston Court Farm

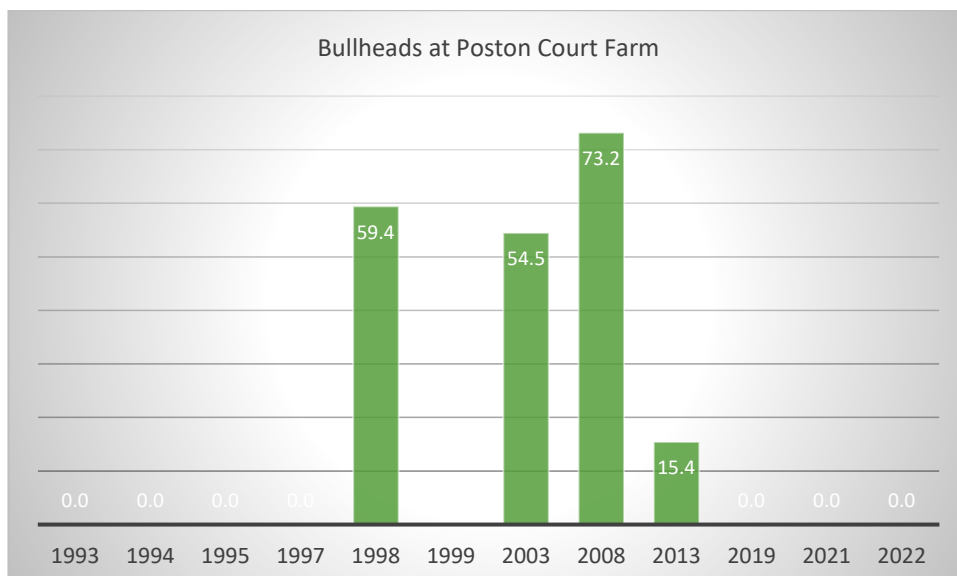


Figure 3: bullhead population (#/100m²) on the Dore at site 10542, Poston Court Farm

Trout are not at the level they were in the early nineties but appear to be hanging in there. Bullheads disappeared after 2013 but neither did they feature in the counts before 1998.

Next, we look further downstream at site 50463, Abbey Dore. Figure 4 presents the data between 2013-19. This shows that during this period, trout numbers fell by almost a factor of 3; bullheads, stone loach and eels disappeared entirely. Taking the combined picture of the roughly simultaneous disappearance

¹ Survey areas are not constant year by year; therefore, the trend charts have been normalised to an area of 100m² – which accounts for the fractions of fish!

the bullhead population at Poston Court Farm paints a picture of a collapsing ecosystem. Thus, the 2022 change of classification for *fish* on the Dore from *high* to *moderate* seems entirely reasonable. The only mystery is why it was classified as *high* in 2019. The only new data would appear to be the continued absence of bullheads at Poston Court Farm.

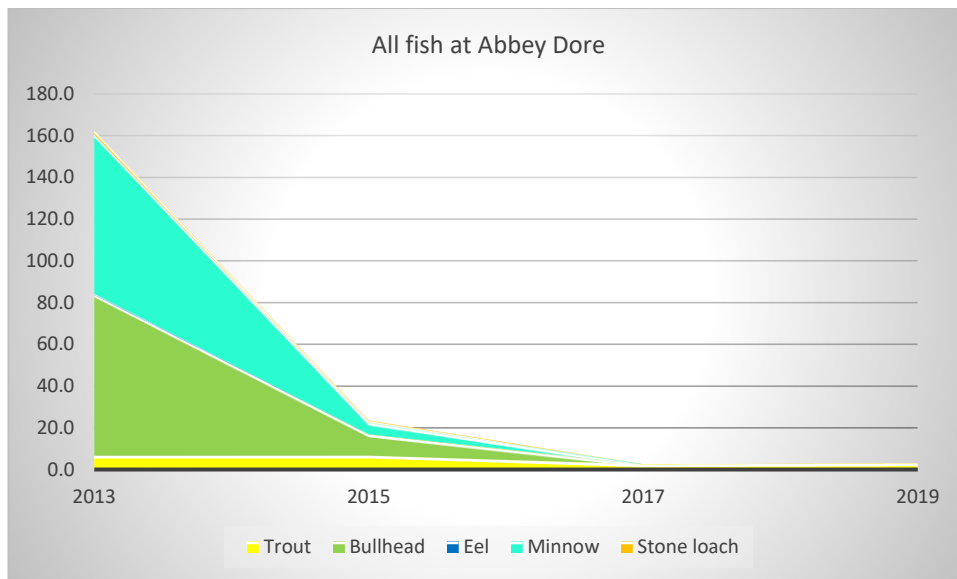


Figure 4: fish population (#/100m²) on the Dore at site 50463, Abbey Dore. Trout fell from 6.6 in 2013 to 3.0 in 2019; stone loach from 1.9 to zero. The European eel is in trouble Europe-wide, so their decline is not a reliable indicator of river health.

Dulas trends

Two surveys have been carried out on the Dulas, both at site 50344, Ewyas Harold. Note that this is well upstream of the large STW at Pontrilas. Data is shown in table 3 below.

Species	2013 (#/100m²)	2019 (~/100m²)
Trout (<i>Salmo trutta</i>)	13.2	4.6
Bullhead (<i>Cottus gobio</i>)	31.0	-
Eel (<i>Anguilla anguilla</i>)	0.8	-
Minnow (<i>Phoxinus phoxinus</i>)	112.6	-

Table 3: fish survey results from the Dulas Brook at site 50344, Ewyas Harold.

Again, we find a complete collapse of bullhead and minnow populations between 2013-19. Trout numbers have declined by almost a factor of 3. Grayling and chub, which might be expected, are completely absent in both years. This is very similar to Abbey Dore, and once again, the reclassification from high to moderate status is entirely understandable. The mystery remains how this could have been classified as high status in 2019, there being no new data since then.

Worm trends

Three surveys have been carried out on the Worm Brook: two in the upper river at Much Dewchurch, and one just above the Dore confluence. Table 4 shows the data at Much Dewchurch.

Species	2008 (#/100m²)	2013 (~/100m²)
Trout (<i>Salmo trutta</i>)	-	-
Bullhead (<i>Cottus gobio</i>)	46.0	20.0
Stickleback (<i>Gasterosteus aculeatus</i>)	3.0	24.0
Minnow (<i>Phoxinus phoxinus</i>)	-	-

Table 4: fish survey results from the Worm Brook at site 29782, Much Dewchurch.

Essentially just a few bullheads and sticklebacks. No minnows, chub, roach, perch, etc. One wonders if the bullheads are still there, given their precipitous collapse on the Dore and the Dulas post-2013. Down at the Dore confluence, we find a wider variety of species, but in conspicuously low numbers.

Species	2013 (#/100m ²)
Trout (<i>Salmo trutta</i>)	1.2
Bullhead (<i>Cottus gobio</i>)	2.8
Eel (<i>Anguilla anguilla</i>)	0.2
Stickleback (<i>Gasterosteus aculeatus</i>)	0.2
Minnow (<i>Phoxinus phoxinus</i>)	28.0
Chub	0.2
Grayling	0.2
Stone loach	2.6

Table 5: fish survey results from the Worm Brook at site 50184, Sunnymede

This data certainly justifies the *moderate* classification in both 2019 and 2022. However, it is surely regrettable that there is no data post-2013. That's ten years without a survey. Things could be even worse now!

Invertebrates

Invertebrate surveys utilise kick sampling. But rather than a field examination, samples are taken back to the laboratory for counting. The variety of organisms logged is extensive, with more granularity appearing in the last few years, presumably a result of improved microscopy techniques. As well as raw counts by taxa, various summary metrics are calculated, with the WHPT index apparently being used in the River Invertebrate Classification Tool (RICT) for WFD compliance. The WHPT index takes account of both variety of species and abundance. The details appear quite complex and would require a lot more digging in order to fully understand how these metrics are generated and used. Below I shall present the WHPT index, in addition to counts for the most common taxa.

There are 17 sites within the Dore sub catchment which have been used for invertebrate sampling, as listed in table 6 and shown on the map in figure 5. Some of these correspond to fish survey sites, presumably because both processes require easy wading access. Key data recorded for each site include substrate composition (% sand, silt, pebbles, boulders), alkalinity & conductivity. All of which would be expected to have a bearing on the range of species living there. Sampling is performed twice, or sometimes three times, within the year.

Table 6 shows us that sites on the Worm, Allensmore and Jury Brooks were substantially terminated during the austerity years. Note that the last survey on the Worm was in 2014 and the last survey on the Dulas was in 2018. Only two sites on the Dore, at Pontrilas and Peterchurch have any long-term continuity of data. So, firstly we shall take a look at these sites.

Waterbody / SiteID	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Allensmore Brook																														
45287																														
River Dore																														
45279																														
45280																														
45642																														
Jury Brook																														
45286																														
158488																														
158489																														
158490																														
Dulas Brook																														
45281																														
158064																														
Worm Brook																														
45282																														
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Table 6: invertebrate surveys within the Dore sub catchment by site and year

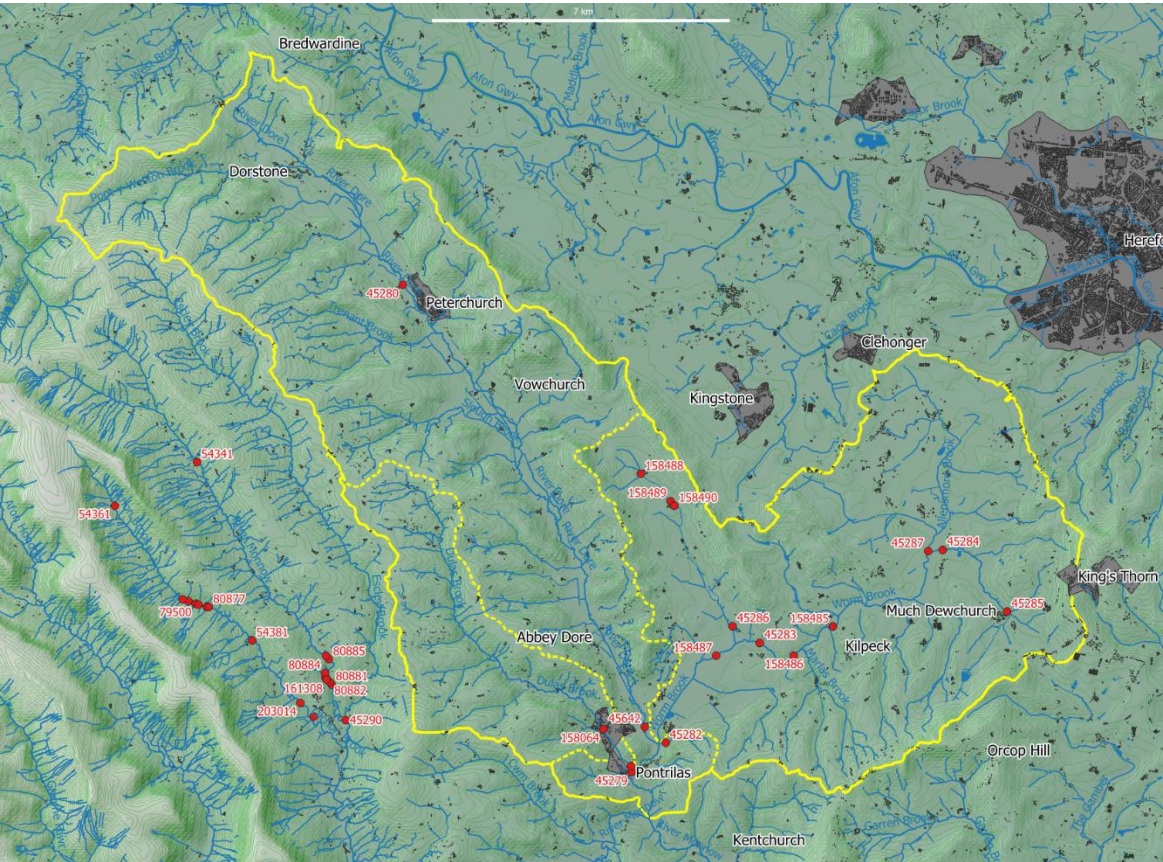


Figure 5: map of the Dore sub catchment showing invertebrate survey sites (red circles).

45280: Dore at Hinton Cross Bridge, Peterchurch

Significantly, note that this location is well upstream of the Peterchurch STW discharge at the south of the village.

The average WHPT index at Hinton Cross Bridge is shown in Figure 6 below. Over the last thirty years, this metric has fluttered around a mean value of 151, with no upward or downward drift. This presumably underpins the constant WFD classification of *high* status.

During 2022, this site was surveyed on three occasions, in April, June and September. Figure 7 shows the relative abundance of taxa, totalled across the three surveys. By far the most abundant invertebrate, making up 84% of the total, is the freshwater shrimp *Gammarus fossarum*.

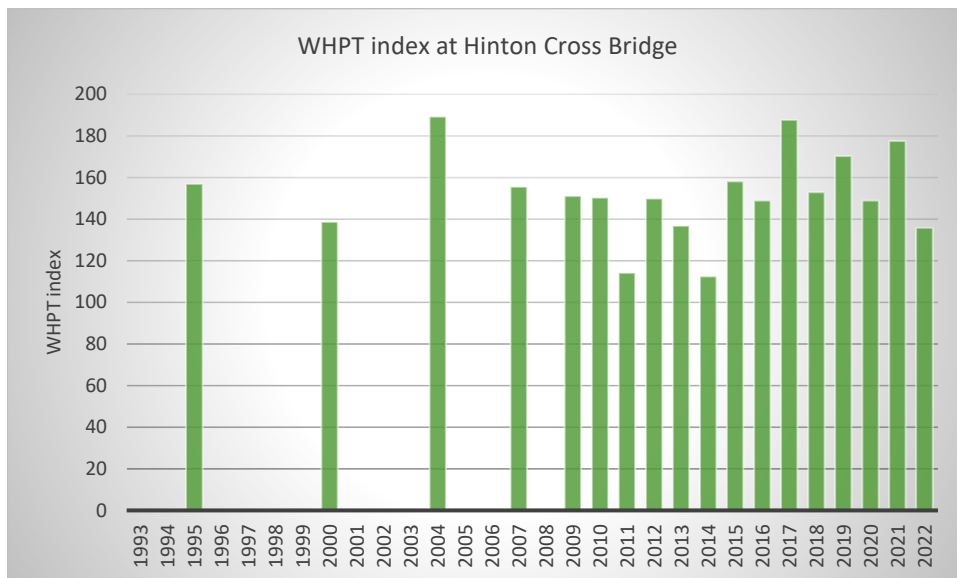


Figure 6: WHPT index (WHPT_TOTAL) on the Dore at site 45280, Hinton Cross Bridge.

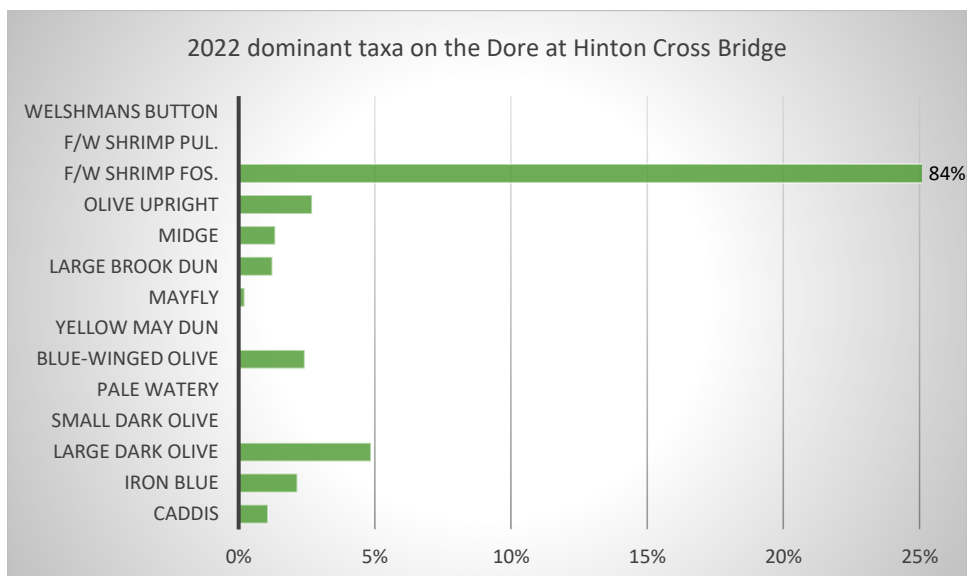


Figure 7: most abundant taxa in 2022, on the Dore at site 45280, Hinton Cross Bridge.

Alongside this crustacean, we find larvae of several of the classic up-winged flies, familiar to trout anglers. We also find chironomid midge larvae and caddis larvae. Mayfly larvae (*Ephemera danica*) are less common, but this is to be expected in a gravel dominated area – this larva prefers silt.

45279: Dore below Pontrilas

Note that this site is below the Worm and Dulas confluences – the latter incorporating Pontrilas STW effluent.

The WHPT trend is shown in Figure 8. The long-term average value is 165 (slightly higher than Hinton Cross Bridge) and has been fairly stable over the last ten years.

This site was also surveyed three times during 2022, in April, June and September. Figure 9 shows the most abundant taxa at this location. *Gammarus fossarum* is still the most abundant invertebrate logged, but here makes up only 37% of the total. In absolute terms, this equates to about a third of the abundance found at Hinton Cross Bridge.

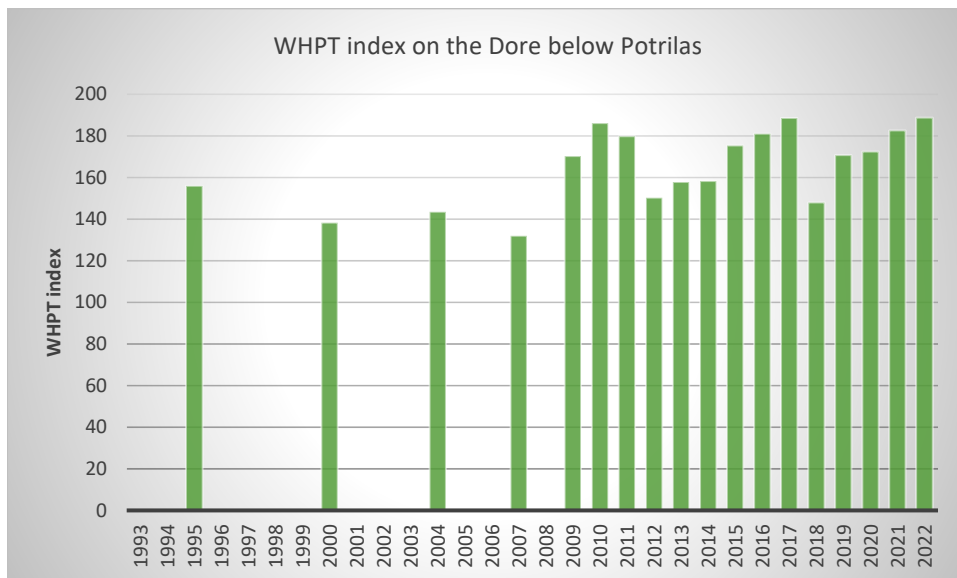


Figure 8: WHPT index (WHPT_TOTAL) on the Dore at site 45279, below Pontrilas.

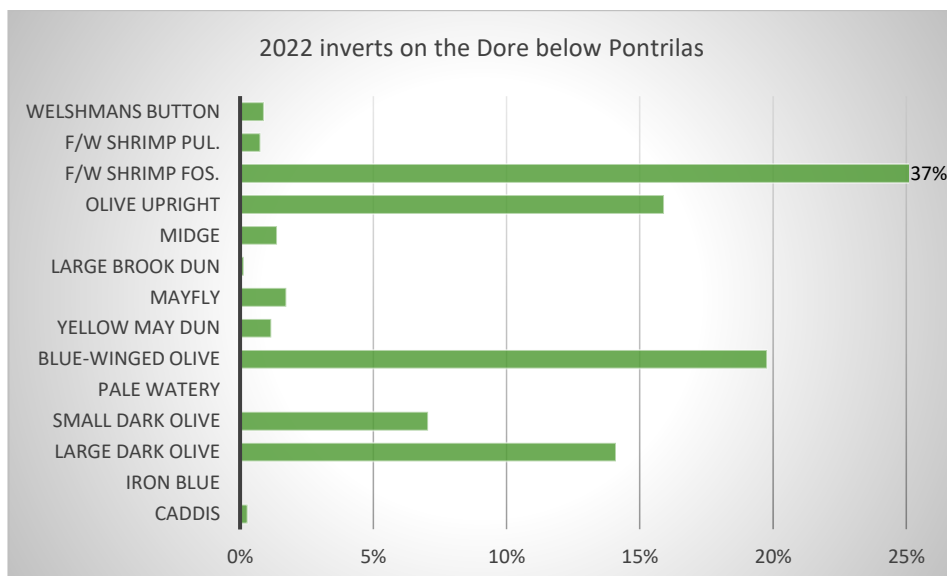


Figure 9: most abundant taxa in 2022, on the Dore at site 45279, below Pontrilas.

Olives are more abundant here than upstream, both in relative and absolute terms. Mayfly larvae (*Ephemera danica*) are also making an appearance, as are yellow may dun larvae (*Heptagenia sulphurea*).

15804: Dulas at Ewyas Harold

Note that this is well upstream of Pontrilas STW. The brook here runs off pasture and has perhaps the lowest phosphate levels in the Dore sub catchment. Sampling on the Dulas has unfortunately been sporadic (see table 6), with this location surveyed only three times since 2013, the most recent being in 2018.

Figure 10 shows the WHPT trend, albeit with only three data points. The average value of 221 is significantly higher than the Dore sites.

Figure 11 shows the dominant taxa recorded in May and October of 2018.

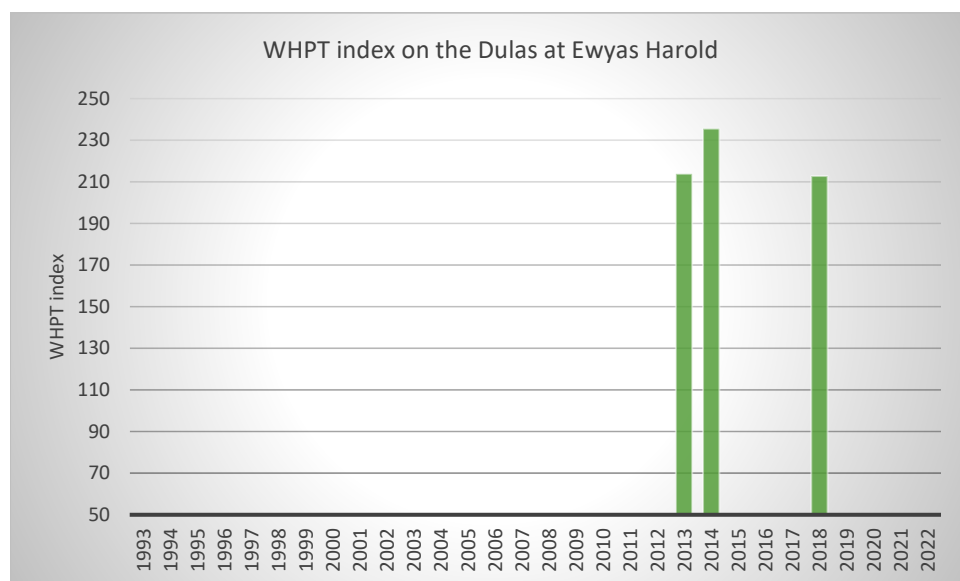


Figure 10: WHPT index (WHPT_TOTAL) on the Dulas at site 15804, Ewyas Harold.

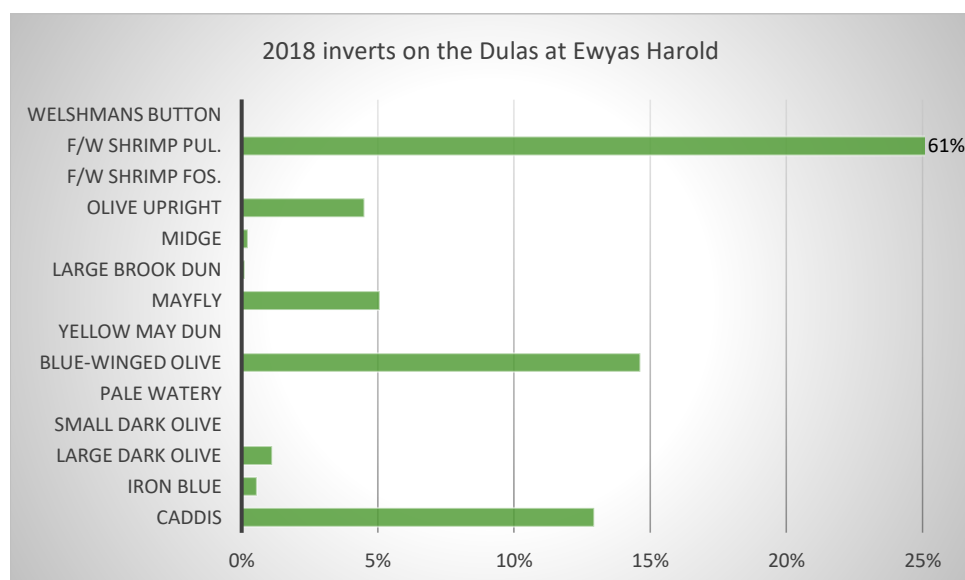


Figure 11: most abundant taxa in 2018, on the Dulas at site 15804, Ewyas Harold.

Once again, a crustacean is top of the table. But this time it is *Gammarus pulex* rather than *Gammarus fossarum*. Beneath this, we find a range of up-winged insects, notably olives, but with mayfly larvae in significant numbers. Caddis larvae are also very significant, but chironomid midges are almost non-existent.

45284: Worm above the Allensmore confluence

This site is located in the upper reaches of the Worm. However, it is the only site on the Worm with any data collected within the last decade, the most recent survey being in 2014. There has been no sampling in the lower reaches of this brook² since the multi-site survey of 2011.

Figure 12 shows the WHPT trend at this site. The average value of 97 is far lower than the Dore and Dulas sites and presumably explains the *good* WFD classification on this waterbody (cf. *high* on the Dore and Dulas).

Figure 13 shows the dominant taxa recorded in 2014. This is entirely different to the Dore and Dulas, with the slower flows of the Worm favouring silt-dwelling species. Gone are the up-winged insects, being replaced by crustaceans, molluscs and worms. The only insects to feature are midges and caddis.

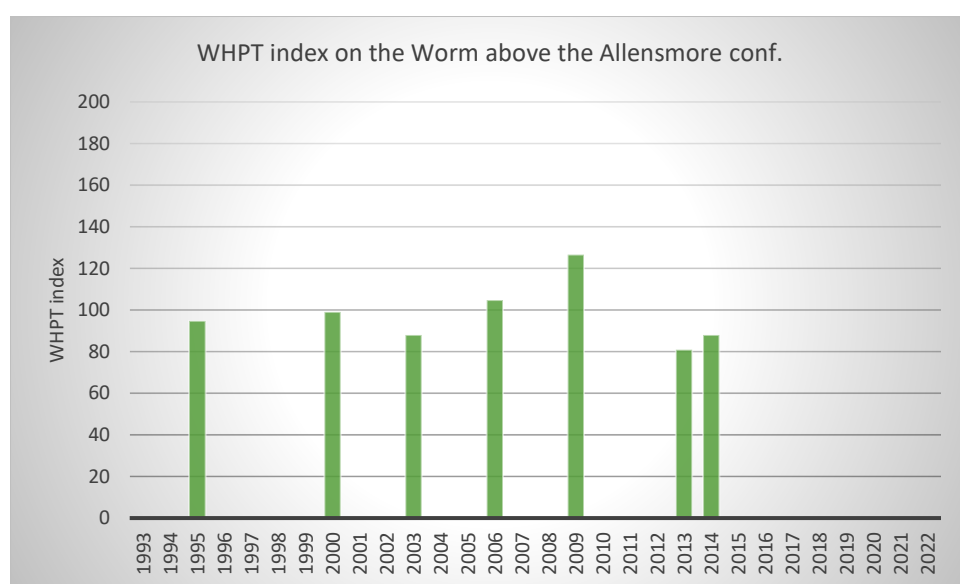


Figure 12: WHPT index (WHPT_TOTAL) on the Worm at site 45284, above the Allensmore conf.

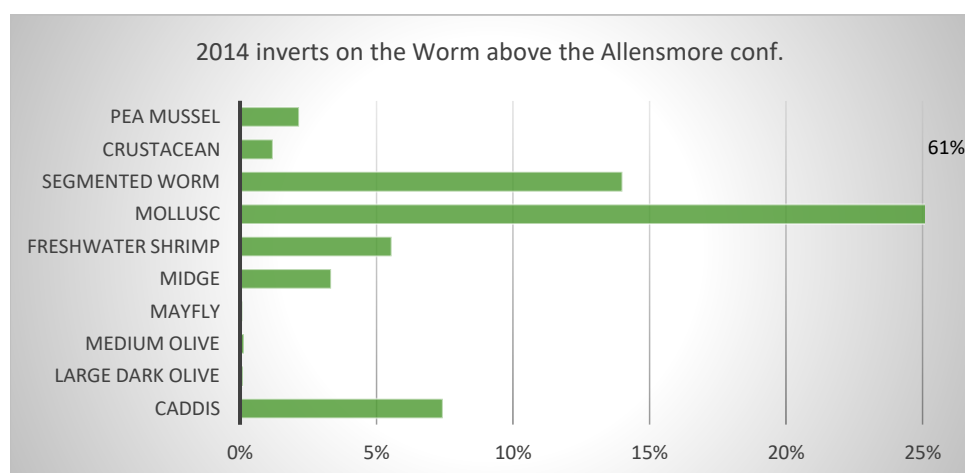


Figure 13: most abundant taxa in 2014, on the Worm at site 45284, above the Allensmore conf.

² Note that site 45282 shown in figure 5 is actually on a side stream and not on the main stem of the Worm.

Macrophytes

Since 2006, EA macrophyte surveys switched to the WFD methodology and are not readily compared to earlier data. As for invertebrates, the macrophyte results are provided both as summary metrics and as underlying taxa abundance data.

Table 4 below shows the survey history for the Dore sub catchment and figure 14 shows the locations of the various survey sites. Surveys are performed once during the summer, in June, July or August.

[illegible]

Table 7: macrophyte surveys within the Dore sub catchment by site and year

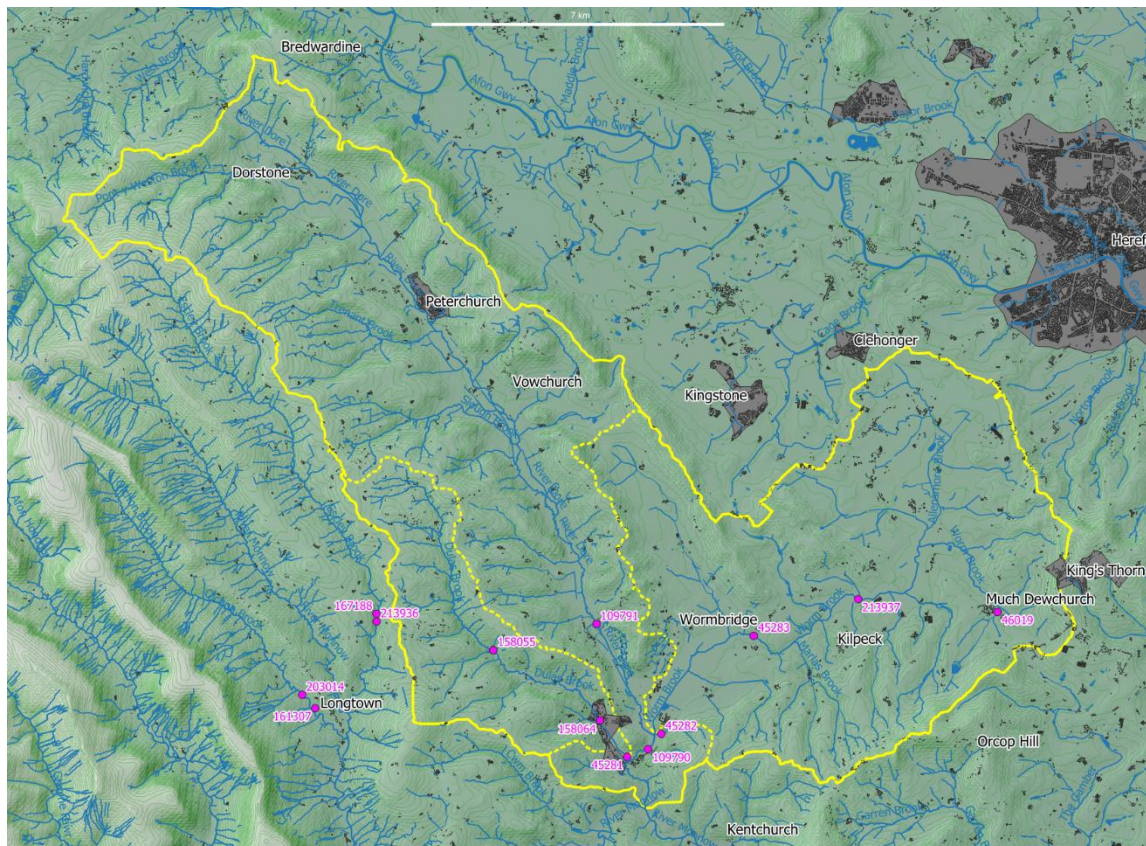


Figure 14: map of the Dore sub catchment showing macrophyte survey sites (pink circles). Note that sites 46019 and 45282 are on tributary streams, not the main stem of the Worm.

No surveys are recorded prior to 2004. Also note that there are no surveys whatsoever on the Dore upstream of Abbey Dore. After 2006 we find the now familiar sporadic pattern emerging, possibly at one location in each waterbody. But this fizzled out entirely in 2014, presumably a victim of austerity. No surveys have been performed since 2014, until one this year which was performed in an entirely new location. This being the case, it is no surprise that WFD classifications for macrophytes have not changed between 2019 and 2022. No one has looked in eight years. Below, we shall examine one site from each waterbody, generally being the only one with any semblance of continuity of surveys (or not, in the case of the Worm).

The dataset is downloaded as two tables. The first table contains summary metrics, the key ones being as follow:

- RMNI (River Macrophyte Nutrient Index): a weighted score (0-10) indicating the preference of macrophyte taxa for nutrient rich water. A low score indicates taxa present prefer low nutrient levels; a high score indicates a preference for high nutrient levels.
- RFA_PC: the percentage cover of filamentous algae.

Both of the above are indicators of eutrophication. The following are indicators of the richness of aquatic taxa:

- N_RFG: the number of functional groups (out of a possible 23) found within the sample.
- RN_A_TAXA: the number of individual aquatic taxa found within the sample.

The second table contains a list of all taxa recorded in each survey. Fractional coverage for each taxon is recorded as a band, rather than a precise number. These are defined as follows:

Coverage band	Fractional coverage
C1	<0.1%
C2	0.1 to 1%
C3	1 to 2.5%
C4	2.5 to 5%
C5	5 to 10%
C6	10 to 25%
C7	25 to 50%
C8	50 to 75%
C9	>75%

Table 8: macrophyte coverage bands

109791: River Dore at Abbey Dore

Two post-2006 surveys are recorded at this site. Metrics from these are summarised in table 9 below.

Year	RMNI	RFA_PC	N_RFG	RN_A_TAXA
2007	6.6	0.1	4	8
2014	7.1	38.0	6	10

Table 9: macrophyte metrics from the River Dore at site 109791, at Abbey Dore.

The standout result here is the dramatic increase in filamentous algae (RFA_PC) between 2007 and 2014. Interestingly, this coincides with the demise of the bullhead population at Abbey Dore. The other numbers are hard to interpret without some wider context.

Individual taxa recorded at Abbey Dore in 2014 are summarised in table 10 below.

Taxon	Taxon group	Cover band	Fractional coverage
Amblystegium	moss	2	0.1 to 1%
Apium nodiflorum	flowering plant	1	<0.1%
Batrachospermum	alga	1	<0.1%
Cladophora	alga	2	0.1 to 1%
Epilobium hirsutum	flowering plant	1	<0.1%
Filamentous green algae	alga	7	25 to 50%
Fontinalis antipyretica	moss	3	1 to 2.5%
Hildenbrandia rivularis	alga	1	<0.1%
Iris pseudacorus	flowering plant	1	<0.1%
Lemna minor	flowering plant	1	<0.1%
Oenanthe crocata	flowering plant	1	<0.1%
Pellia endiviifolia	liverwort	2	0.1 to 1%
Ranunculus (Batrachian) spp.	flowering plant	1	<0.1%
Ranunculus repens	flowering plant	1	<0.1%
Rhynchostegium	moss	2	0.1 to 1%
Scrophularia auriculata	flowering plant	1	<0.1%

Table 10: macrophyte taxon coverage from the River Dore at site 109791, at Abbey Dore, in 2014.

These results underline the high RFA_PC score. Filamentous algae are covering a major fraction of the riverbed (Cladophora is also a filamentous alga). Beyond this, very little grows. Flowering plants are present only in tiny amounts.

158064: Dulas Brook at Ewyas Harold

Two post-2006 surveys are also recorded at Ewyas Harold. Metrics are summarised in table 11 below.

Year	RMNI	RFA_PC	N_RFG	RN_A_TAXA
2011	7.8	42.5	5	9
2014	7.4	4.3	4	7

Table 11: macrophyte survey results from the Dulas Brook at site 158064, at Ewyas Harold.

Here we find the opposite change on RFA_PC, the Dulas being cleaner in 2014. The other metrics look broadly similar to the Dore values.

Individual taxa recorded at Ewyas Harold in 2014 are listed in table 12 below.

Taxon	Taxon group	Cover band	Fractional coverage
Amblystegium	moss	1	<0.1%
Apium nodiflorum	flowering plant	1	<0.1%
Brachythecium	moss	1	<0.1%
Catabrosa aquatica	flowering plant	1	<0.1%
Cladophora	alga	2	0.1 to 1%
Conocephalum conicum	liverwort	1	<0.1%
Epilobium hirsutum	flowering plant	1	<0.1%
Filamentous green algae	alga	4	2.5 to 5%
Hildenbrandia rivularis	alga	1	<0.1%
Oenanthe crocata	flowering plant	1	<0.1%
Phalaris arundinacea	flowering plant	1	<0.1%
Ranunculus repens	flowering plant	1	<0.1%
Rhynchostegium riparioides	moss	1	<0.1%
Rumex	flowering plant	1	<0.1%

Table 12: macrophyte taxon coverage from the Dulas Brook at site 158064, at Ewyas Harold, in 2014.

The range of taxa is almost the same. Again, only filamentous algae are present in any significant amount. The coverage of flowering plants is vanishingly small.

45283: Worm Brook at Wormbridge

There is only one post-2006 survey recorded for the Worm (recall that sites 45282 and 46019 are on side streams). Metric results are shown in table 13 below.

Year	RMNI	RFA_PC	N_RFG	RN_A_TAXA
2007	7.6	38.1	5	11

Table 13: macrophyte survey results from the Worm Brook at site 45283, at Wormbridge.

On the Worm, we find heavy filamentous algal cover present several years earlier than on the Dore and Dulas (admittedly based on a single data point). The other metrics are not dissimilar to the Dore and Dulas values.

Individual taxa recorded at Wormbridge in 2007 are listed in table 14 below.

Taxon	Taxon group	Cover band	Fractional coverage
<i>Alisma plantago-aquatica</i>	flowering plant	1	<0.1%
<i>Amblystegium fluviatile</i>	moss	1	<0.1%
<i>Apium nodiflorum</i>	flowering plant	1	<0.1%
<i>Cladophora</i>	alga	2	0.1 to 1%
<i>Enteromorpha intestinalis</i>	alga	1	<0.1%
<i>Fissidens</i>	moss	1	<0.1%
<i>Lemanea</i>	alga	1	<0.1%
<i>Lemna minor</i>	flowering plant	1	<0.1%
<i>Leptodictyum riparium</i>	moss	2	0.1 to 1%
<i>Myosotis</i>	flowering plant	1	<0.1%
<i>Oenanthe crocata</i>	flowering plant	2	0.1 to 1%
<i>Pellia endiviifolia</i>	liverwort	2	0.1 to 1%
<i>Persicaria hydropiper</i>	flowering plant	1	<0.1%
<i>Phalaris arundinacea</i>	flowering plant	2	0.1 to 1%
<i>Rhynchostegium riparioides</i>	moss	1	<0.1%
<i>Rorippa nasturtium-aquaticum</i>	flowering plant	1	<0.1%
<i>Sparganium erectum</i>	flowering plant	5	5 to 10%
<i>Vaucheria</i>	alga	7	25 to 50%

Table 14: macrophyte taxon coverage from the Worm Brook at site 45283, at Wormbridge, in 2007.

The range of taxa is a little different here, presumably as a result of the more leisurely flowrates and siltier bed found on the Worm. But, again, other than *Sparganium* (branched bur reed, found along the margins), flowering plants are struggling. *Vaucheria*, an alga which is otherwise known as water felt, is the dominant taxon present.

Summary and conclusion

Fish

- WFD classification was downgraded from *high* in 2019 to *moderate* in 2022 for both the Dore and Dulas, with the Worm classified as *moderate* in both assessments.
- Fish surveys in recent years have been infrequent. The Worm was last surveyed in 2013, the Dulas in 2019.
- The Dore at Poston Court Farm, below Peterchurch, has been more regularly surveyed. Results show brown trout at less than half of the population present in the mid-nineties. The bullhead population collapse here around 2013. But equally, there were no bullheads prior to 1998.
- The Dore at Abbey Dore has been surveyed four times between 2013 and 2019. Results show a massive collapse in minnows and bullheads over the period 2013-2017. Trout numbers also declined by a factor of two.
- The Dulas at Ewyas Harold was surveyed in 2013 and 2019. Between these dates, there was a total collapse of minnow and bullhead populations. Trout numbers fell by a factor of three.

Invertebrates

- The Dore and Dulas waterbodies have been assessed as *high* in both 2019 and 2022. The Worm is also unchanged, with a *good* assessment.
- Invertebrate surveys have been infrequent in recent years. The Worm was last surveyed in 2014 and the Dulas in 2018. Two sites on the Dore are the only locations to have had regular surveys.
- Trends on WHPT index appear to be stable on all three waterbodies over the period of available data. But the lack of any recent surveys on the Worm and Dulas is of concern.
- Results presented for taxa abundance provide some guidance on what to expect in any future CS invertebrate sampling. Note that the Worm is radically different to the Dore and Dulas in this respect.

Macrophytes

- In both 2019 and 2022, the Dore and Worm were classified as moderate for macrophytes. The Dulas was classified as good.
- Macrophyte surveys have not been carried out in any of the three waterbodies since 2014, until this year.
- In the most recent surveys (2014), the Dore and Worm are dominated by filamentous algal cover. In the 2007 Dore survey, this was not the case. A range of flowering plants are present at both sites but at very low coverage levels.
- The Dulas in 2014 had much lower coverage of filamentous algae than the Dore and Worm. Presumably this is why it achieves a higher WFD classification. The range of flowering plants is otherwise similar to the Dore, and with similar, low coverages.
- Results presented for taxa coverage provide guidance for any future CS macrophyte monitoring.

Whilst the above is far from a comprehensive review – and as a non-biologist, I'm not sure I'm competent to do that in any event – I think this gives us a feel for what to expect in terms of types of invertebrates and macrophytes might be encountered in any future CS work.

More generally, it tells a tale of three struggling waterbodies, as acknowledged by the EA's 2022 WFD classifications for fish and macrophytes. The Worm is a basket case, and the Dore and Dulas are in decline. There is an interesting correlation between the appearance of heavy filamentous algal coverage and the demise of bullheads and minnows on the Dore, backed up by a similar pattern on the Dulas. Something happened, roughly over the period 2010-20. However, we do need to be a little careful insofar as a correlation doesn't prove a causal relationship. As well as water quality issues, there could be predation or disease at play. But the simultaneous collapses are certainly suspicious.

The consistent high statuses awarded for *invertebrates* flies in the face of the situation for fish and macrophytes. I have a concern that the metrics used to judge this, being partly based on variety of taxa, are not considering of whether they are the right taxa. The absence of up-wing flies and high abundance of molluscs and worms on the Worm seems to me (as a non-biologist) to be an example of a high-scoring but entirely degraded ecosystem.

The paucity of recent surveys in all three categories is a cause of great concern. The impact of reduced resourcing at the Environment Agency since 2010 is clear for all to see. Without suggesting that the recent WFD classifications are incorrect, it is nevertheless hard to escape the conclusion that their evidence base is somewhat lacking.

List of abbreviations

CS	Citizen Science
RICT	River Invertebrate Classification Tool (used for WFD compliance)
WHPT	Walley Hawkes Paisley Trigg index (used in RICT)
WFD	Water Framework Directive
WSA	Wye Salmon Association

References